

**CO4 – AT2 -INDUSTRY PROBLEM
SOLVING**

**Course Title: Machine Learning Course
Code: ITA0609**

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1. A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity between users is measured, how recommendations are generated, and how the system adapts to changing user behavior over time. ?.

1. Problem Definition

- The company wants to recommend the **most suitable data plan** based on individual customer usage.
- Customer behavior can be represented using **monthly data usage, call duration, SMS usage, and streaming hours**.
- **KNN** is selected because customers with similar usage patterns are likely to require similar data plans.

2. User Data and Features

The system collects the following features:

- **Data usage:** GB/month
- **Call duration:** minutes/month
- **SMS usage:** messages/month
- **Streaming:** hours/month

A new customer's usage profile is compared with existing customers whose plans are already known.

3. Measuring Similarity

KNN identifies similar customers using a **distance measure**.

The commonly used Euclidean distance is:

$$[d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2}]$$

A **smaller distance** means greater similarity between two customers.

Before calculating distance, features should be **normalized** because data usage and call minutes have different numerical scales.

4. Numerical Example

Consider a new customer:

New Customer = (20 GB, 300 minutes)

Three existing customers are:

Customer A = (18 GB, 280 minutes)

Customer B = (22 GB, 310 minutes)

Customer C = (40 GB, 500 minutes)

For Customer A:

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$$[d_A = \sqrt{(20-18)^2 + (300-280)^2}]$$

$$[d_A = \sqrt{4+400}]$$

$$[d_A = \sqrt{404}]$$

$$[\boxed{d_A = 20.10}]$$

For Customer B:

$$[d_B = \sqrt{(20-22)^2 + (300-310)^2}]$$

$$[d_B = \sqrt{4+100}]$$

$$[\boxed{d_B = 10.20}]$$

For Customer C:

$$[d_C = \sqrt{(20-40)^2 + (300-500)^2}]$$

$$[d_C = \sqrt{400+40000}]$$

$$[\boxed{d_C = 201.00}]$$

Therefore:

$$[d_B < d_A < d_C]$$

Customer B is the most similar customer because it has the smallest distance.

5. Generating the Recommendation

The system selects the **K nearest customers** and examines the plans used by them.

For example, if **K = 3** and the three nearest customers have:

- Customer B → **Standard Plan**
- Customer A → **Standard Plan**
- Customer D → **Premium Plan**

Then:

Standard Plan = 2 votes

Premium Plan = 1 vote

Therefore, the system recommends the **Standard Plan**.

This is called **majority voting**, where the most common plan among the nearest customers becomes the recommendation.

6. Analytical Decision

The recommendation should also consider whether the customer's usage is close to the **plan limit**.

For example, suppose:

Standard Plan limit = 25 GB

New customer's usage:

20 GB/month

Usage percentage:

$$[\text{Usage\%} = \frac{20}{25} \times 100]$$

$$[\boxed{80\%}]$$

The customer is using **80% of the available data**, so the Standard Plan may be suitable but should be monitored.

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If usage increases to **30 GB/month**, the customer exceeds the 25 GB limit.
Therefore, the system can recommend a **Premium Plan** with a higher data allowance.

7. Adapting to Changing User Behaviour

Customer usage can change over time because of **streaming, work, travel, gaming, or changes in internet usage**.

The system continuously updates the customer's recent usage profile.

For example:

Previous usage = 5 GB/month

Current usage = 30 GB/month

The customer's nearest neighbours may therefore change from **Basic Plan users to Premium Plan users**.

KNN can adapt by:

- Adding **new usage records**.
- Giving more importance to **recent usage**.
- Updating the customer's feature values.
- Recalculating the nearest customers.
- Generating a new recommendation when the usage pattern changes.

8. Benefits of the Proposed Solution

- **Personalized:** Recommendations are based on individual usage.
- **Simple:** KNN is easy to implement and understand.
- **Adaptive:** New customer data can be incorporated continuously.
- **Multi-feature:** Several usage parameters can be considered.
- **Customer-friendly:** Helps avoid unsuitable or insufficient plans.
- **Business benefit:** Can improve **customer satisfaction and plan selection**.

9. Conclusion

- A **K-Nearest Neighbour based recommendation system** can recommend telecom data plans by comparing a customer's usage with similar existing customers.
- **Euclidean distance** measures similarity, while **KNN majority voting** generates the recommendation.

By continuously updating usage data, the system can detect **changing customer behaviour** and recommend a more suitable plan over time.

2. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data ?.

1. Problem Definition

- The company wants to predict **electricity consumption** using factors such as **time, weather, and appliance usage**.
- Electricity demand changes according to **hour, temperature, rainfall, and household activities**.
- **Locally Weighted Regression (LWR)** is suitable because it gives greater importance to data points that are **similar to the current situation**.

2. Input Features

The model can use:

- **Time:** Hour of the day
- **Temperature:** °C
- **Appliance usage:** Number of active appliances
- **Weather:** Normal, rainy, or extreme conditions
- **Target:** Electricity consumption in kWh

For example:

Input = (Time = 18:00, Temperature = 30°C, Appliances = 6)

The model predicts the expected **electricity consumption**.

3. Local Learning

Unlike ordinary regression, LWR does not give equal importance to every historical observation. It gives a higher **weight** to observations that are closer to the current input.

A Gaussian weighting function can be written as:

$$[w_i = e^{-\frac{(x - x_i)^2}{2\tau^2}}]$$

Where:

(w_i) = weight of data point (i)

(x) = current condition

(x_i) = historical condition

(τ) = bandwidth controlling how quickly the weight decreases

A **higher weight** means the historical observation is more relevant to the current situation.

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4. Numerical Example

Suppose the current temperature is:

30°C

Three historical observations are:

Day 1 = 29°C, consumption = 8 kWh

Day 2 = 30°C, consumption = 9 kWh

Day 3 = 35°C, consumption = 13 kWh

Assume:

$[\tau=5]$

For Day 1:

$[w_1=e^{-\frac{(30-29)^2}{2(5)^2}}]$

$[w_1=e^{-0.02}]$

$[\boxed{w_1=0.98}]$

For Day 2:

$[w_2=e^{-\frac{(30-30)^2}{2(5)^2}}]$

$[\boxed{w_2=1.00}]$

For Day 3:

$[w_3=e^{-\frac{(30-35)^2}{2(5)^2}}]$

$[w_3=e^{-0.50}]$

$[\boxed{w_3=0.61}]$

Therefore, the observations at **29°C and 30°C** receive **higher weights** than the 35°C observation.

5. Predicted Consumption

A simple weighted prediction can be calculated as:

$[\hat{y}=\frac{\sum w_i y_i}{\sum w_i}]$

Substituting the values:

$[\hat{y}=\frac{(0.98 \times 8)+(1.00 \times 9)+(0.61 \times 13)}{0.98+1.00+0.61}]$

$[\frac{7.84+9+7.93}{2.59}]$

$[=\frac{24.77}{2.59}]$

$[\boxed{\hat{y}=9.56 \text{ kWh}}]$

Result: The predicted electricity consumption is approximately **9.56 kWh**.

6. Capturing Local Consumption Variations

Electricity usage can be different during different periods.

For example:

- **Morning:** Lower appliance usage.
- **Afternoon:** Consumption may increase due to cooling systems.
- **Evening:** Higher usage due to lighting, cooking, and entertainment.
- **Night:** Consumption generally decreases.

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LWR focuses on **nearby historical patterns**, so the prediction can change according to the current time and conditions.

7. Handling Noisy Data

Electricity measurements may contain noise because of:

- **Sensor errors**
- Sudden appliance switching
- Communication errors
- Unusual household activities

LWR reduces the effect of unrelated observations because distant data points receive **lower weights**.

For example, if a normal evening consumes around **10 kWh** but one day records **20 kWh** due to an unusual event, that observation should have less influence when predicting a normal evening.

8. Handling Seasonal Data

Electricity consumption changes between seasons.

For example:

- **Summer:** Higher cooling and air-conditioner usage.
- **Winter:** Higher heating or hot-water usage.
- **Rainy season:** Weather-dependent appliance usage may change.

The model can use **recent and seasonally similar observations** to produce more relevant predictions.

Thus, local learning captures changes that a single global regression model may not represent effectively.

9. Advantages

- **Local adaptation:** Focuses on similar historical conditions.
- **Seasonal awareness:** Can capture changing consumption patterns.
- **Noise handling:** Distant observations have less influence.
- **Flexible:** Can model non-linear consumption behavior.
- **Real-time prediction:** Can use newly collected electricity data.

10. Conclusion

Locally Weighted Regression is suitable for electricity consumption prediction because it gives greater importance to historical observations similar to the current conditions.

In the numerical example, the predicted consumption is **9.56 kWh**. By using **local weighting**, the system can capture time-based, weather-based, and seasonal variations while reducing the influence of irrelevant or noisy observations.